COLLECTION OF CTL DECISIONS

Decision

Standard: IEC 60950 (1991) 2nd ed., Amdts.1 (1992), 2 (1993), 3 (1995), 4 (1996)	**Sub clause:** 5.4.6	**Sheet n.** 390 **Page** 1(1)
Subject: Abnormal test	**Key words:** - Abnormal test - Fault	**Decision confirmed at CTL 38th meeting**

Question:

If a printed wiring board mains trace opens during the performance of the component abnormal test, should the test be terminated, should the trace be shorted and the test be continued until the ultimate results are obtained, or should the test be repeated a total of three times?

Decision:

For primary circuits, the trace should be shorted and the test continued until the ultimate results are obtained, additional short circuiting applied as required. For secondary circuits, where the trace is designed to be opened in a repeatable manner, a second option is for the test to be repeated three times to determine the circuit opens repeatedly.

Explanatory notes:

The opening of the trace to terminate an abnormal operating condition cannot be considered the same way as the opening of a fuse or circuit breaker to terminate the same condition. In the primary circuit, if it can be shown that trace that opens is not directly relied upon to protect the entire circuit, then, there is no compromise in safety.
In the secondary circuit, if a trace can be demonstrated to open in a repeatable manner, then, again there is no compromise in safety.